

(22)

$$A = \begin{pmatrix} 1 & 0 & -1 \\ 2 & \lambda & -1 \\ \lambda+1 & 1 & \lambda \end{pmatrix} \text{ とする.}$$

$$A\vec{x} = \vec{0} \text{ かつ } \vec{x} = \vec{0} \text{ の解を求めよ}$$

$$\Leftrightarrow \det A = 0$$

$$\det A = (2\lambda - 1)(\lambda + 1)$$

$$\therefore \lambda = -1, \frac{1}{2}$$

(23)

$$\begin{vmatrix} 2 & 3 & 1 & 0 \\ 3 & 0 & 1 & 0 \\ 2 & 0 & 4 & -1 \\ 1 & 0 & 1 & 2 \end{vmatrix} = - \begin{vmatrix} 3 & 2 & 1 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 2 & 4 & -1 \\ 0 & 1 & 1 & 2 \end{vmatrix}$$

$$= -3 \begin{vmatrix} 3 & 1 & 0 \\ 2 & 4 & -1 \\ 1 & 1 & 2 \end{vmatrix}$$

$$= -3 \cdot 3 \begin{vmatrix} 4 & -1 \\ 1 & 2 \end{vmatrix} + 3 \begin{vmatrix} 2 & -1 \\ 1 & 2 \end{vmatrix}$$

$$= -9(8 + 1) + 3(4 + 1)$$

$$= -81 + 15$$

$$= -66$$

(2)

$$\begin{vmatrix} 3 & 5 & 7 & 0 & -1 \\ 2 & 1 & 6 & 1 & 0 \\ 1 & 6 & 7 & 2 & 2 \\ 6 & 0 & 3 & 0 & 5 \\ 0 & 0 & 5 & 1 & -1 \end{vmatrix} = - \begin{vmatrix} 5 & 3 & 7 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 \\ 0 & 6 & 1 & 2 & 2 \\ 0 & 0 & 3 & 0 & 5 \\ 0 & 0 & 5 & 1 & -1 \end{vmatrix}$$

$$= -5 \begin{vmatrix} 2 & 0 & 1 & 0 \\ 1 & 7 & 2 & 2 \\ 0 & 3 & 0 & 5 \\ 0 & 5 & 1 & -1 \end{vmatrix} + \begin{vmatrix} 3 & 7 & 0 & -1 \\ 1 & 7 & 2 & 2 \\ 0 & 3 & 0 & 5 \\ 0 & 5 & 1 & -1 \end{vmatrix}$$

$$= -5 \times 2 \begin{vmatrix} 7 & 2 & 2 \\ 3 & 0 & 5 \\ 5 & 1 & -1 \end{vmatrix} + 5 \begin{vmatrix} 0 & 1 & 0 \\ 3 & 0 & 5 \\ 5 & 1 & -1 \end{vmatrix}$$

$$+ 3 \begin{vmatrix} 7 & 2 & 2 \\ 3 & 0 & 5 \\ 5 & 1 & -1 \end{vmatrix} - \begin{vmatrix} 7 & 0 & -1 \\ 3 & 0 & 5 \\ 5 & 1 & -1 \end{vmatrix}$$

$$= -10(50 + 6 - 35 + 6)$$

$$+ 5(25 + 3)$$

$$+ 3(50 + 6 - 35 + 6)$$

$$- (-3 - 35)$$

$$= -7 \times (27) + 5 \times 28 + 38$$

$$= -189 + 140 + 38$$

$$= -189 + 178 = -11$$

$$\boxed{24} \begin{vmatrix} a_0 - 1 & 0 & \cdots & 0 \\ a_1 & x & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_n & \cdots & \cdots & x \end{vmatrix} = a_0 \begin{vmatrix} x & 0 & \cdots & 0 \\ 0 & x & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \cdots & x \end{vmatrix} + \begin{vmatrix} a_1 & 1 & \cdots & 0 \\ a_2 & x & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_n & \cdots & \cdots & x \end{vmatrix}$$

$$= a_0 x^n + (a_1 x^{n-1} + \cdots + a_n)$$

$\boxed{25}$   $A$  の余因子行列の成分は  $A$  の元 (整数) の和・差・乗で表わされる。よって  $\hat{A}$  の成分は整数 ( $A$  整数行列  $\Rightarrow \hat{A}$  整数)

よって  $A^{-1} = \frac{1}{\det A} \hat{A}$  ( $\hat{A}$  整数)  $\hat{A} = \det A \cdot A^{-1}$  ( $\det A = 1$ )

3.9.4 (1)  $\hat{A} = (\det A) \cdot A^{-1} \Rightarrow \det \hat{A} = (\det A)^{n-1}$

(2)  $B = \hat{A}$  とおく

$$\tilde{B} = (\det B) \cdot B^{-1} = (\det A)^{n-1} \cdot \frac{A}{\det A}$$

$$\therefore \hat{A} \left( \frac{1}{\det A} A \right) = E_n \quad \hat{A} = (\det A)^{n-2} A$$

- 22, 23, 24, 25
- 3, 4, 4 (1)  $\neq$  (2)  $\neq \det A \neq 0$  (2)
- 3, 5, 1 ~ 3, 5, 3 (時間あれば)

(もしわからない場合  
3, 4, 1 (1), 3, 4, 2 (1), (2), (3))